

STEERABLE VIDEO-TO-AMBISONICS SPATIALIZATION: FAITHFUL DIRECTION CONTROL AND HOW PRECISE IT NEEDS TO BE

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ABSTRACT

Video-to-ambisonics models for 360-degree video are conditioned on a source direction, an oracle on the YT-Ambigen benchmark. Deployed, the direction comes from a user or a localizer, so the model should treat it as a *control*: put the sound where it says, anywhere on the sphere, and degrade gracefully when it is imprecise. Neither property has been measured. We propose a steerability protocol built on the exact rotation group of first-order ambisonics, which needs no annotation: an equivariance test, a follow test with random targets, and a control-precision curve. A spatializer trained on ground-truth directions is only approximately steerable: its peak follows a random target in 91% of the clips, but its field turns diffuse away from the training data’s frontal directions. *Rotation-consistency training*, a free augmentation, makes the control exact (100% of targets, median 3°) and raises the benchmark score from CC 0.783 to 0.805. The precision curve of a faithful model is a data-only bound scaled by its accuracy: 30° of error costs 0.13 CC, a click on a 7×14 grid 0.015, and beyond 47° a direction is worse than none, so a leak-free panoramic localizer with 58° median error does not help.

Index Terms— Spatial audio, first-order ambisonics, 360-degree video, controllable generation, sound source localization

1. INTRODUCTION

A sound track for a 360-degree video must not only sound right, it must *arrive from the right direction*. First-order ambisonics (FOA) generators, whether end-to-end [1, 2, 3, 4] or two-stage systems that generate a mono track and then spatialize it [5], therefore take a source direction as input. On the YT-Ambigen benchmark [1] this direction is the “camera direction” shipped with every clip, and we showed in [6] that it is an oracle: it coincides with the arg-max of the ground-truth energy map, a plane wave placed at it already scores CC 0.69, and the same spatializers fall from CC 0.78 to 0.50 once it is withheld.

The oracle is not available in practice, but the direction input is still the natural interface of such a system: a sound designer points at the panorama, a tracker follows a character, an audio-visual localizer proposes a direction. All these uses assume two properties

that nobody has measured. *Fidelity*: when the direction disagrees with what the video and the audio suggest, does the model obey the direction? A network trained only on ground-truth directions has never seen a disagreement and is free to rely on the frontal bias of the data instead. *Tolerance*: a user or a localizer is imprecise, so how much spatial quality is lost per degree of error, and at what error is the direction worse than none?

This paper answers both questions for the spatialization setting of [6, 5] (mono W and field-of-view video given, predict the directional channels), with three contributions. (i) A *steerability protocol* (Sec. 3) derived from the exact rotational closure of first-order ambisonics, which needs no annotation: rotating the direction input about the vertical axis must rotate the produced spatial image by the same angle, which can be scored with the official energy-map metrics against the rotated ground truth; uniform random targets measure whether the model can place a sound anywhere; and perturbing the direction by a known angle traces the control-precision curve. (ii) *Rotation-consistency training* (Sec. 3), a free augmentation that rotates the target field and the direction input together. The standard recipe turns out to be only approximately steerable: the peak follows a random target in 91% of the clips, but the field loses its contrast away from the frontal directions of the training data. With $SO(3)$ rotation consistency the control becomes exact (100% of targets, median 3°, equivariance to three decimals) and the benchmark score rises from CC 0.783 to 0.805 (Sec. 5.1). (iii) *Design findings*: the precision curve of a faithful spatializer is a data-only bound scaled by its accuracy, so 30° of error costs 0.13 CC, a click on a 7×14 equirectangular grid retains 95% of the oracle gain, and directions with errors beyond $\approx 47^\circ$ are worse than none; training with noisy directions buys tolerance only by giving up fidelity. Finally, we train a leak-free panoramic audio-visual localizer (dense CLIP tokens of the equirectangular frame, tagged with their directions, attended by an audio-derived query and supervised by the ground-truth energy map) as an automatic control; its 58° median error lands where the curve predicts, below the no-direction baseline, and we show why localizing sound on YouTube panoramas is harder than the benchmark suggests.

Related work. Visually guided upmixing of mono audio to binaural or ambisonic formats regresses the missing channels from video [7, 8, 9]; spatial audio-visual alignment has served as a self-supervised pretext [10]. Audio-visual source localization attends from audio to image regions [11, 12]; our localizer does so on the sphere with the FOA energy map as a free dense target. Sound-field rotation is used in [1] as a single 90° augmentation and by

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DynFOA [3] to follow the viewing direction; we use the continuous group as a training signal and, new to our knowledge, as a *test* of controllability. Controllability of audio generators is usually assessed for text or timing conditions [13], never for the direction condition.

2. SETTING

Task, data, metrics. A clip provides the mono channel W , a field-of-view (FoV) video V and the target channels (Y, Z, X) in ACN/SN3D order W, Y, Z, X ; the model f predicts $(\hat{Y}, \hat{Z}, \hat{X}) = f(W, V, \mathbf{d})$ from W, V and a unit direction $\mathbf{d}$. We use the retrievable YT-Ambigen subset of [6] (48,512 / 6,122 / 6,553 train / validation / test clips of 5 s, video-disjoint, audio at 16 kHz) and its official evaluator [1]: the field is decoded on a 2° spherical grid by the first-order harmonics $\mathbf{h}(\mathbf{u}) = (1, u_y, u_z, u_x)$, the RMS map $E(\mathbf{u}) = (\mathbf{h}(\mathbf{u})^\top Q \mathbf{h}(\mathbf{u}))^{1/2}$ with Q the 4×4 covariance of the window is computed over the clip (“all”), at 1 fps and 5 fps, equal-area resampled and scored by CC and AUC [14] against the ground truth. We report CC and AUC of the full clip and CC at 1 fps; the 5 fps numbers follow the same ordering.

Spatializer. f is the two-level complex-mask U-Net of [6] (8.4M parameters): the STFT of W (1024/256) enters as real, imaginary and log-magnitude planes, every block is FiLM-modulated by a conditioning sequence made of CLIP ViT-B/16 features of V at 4 fps [15], an MLP embedding of $\mathbf{d}$ and a learned temporal position, and the output is a complex ratio mask per directional channel. It is trained with the metric-aligned objective of [6], the normalized complex MSE plus $\lambda = 3$ times one minus the correlation of decoded energy maps, which we keep unchanged. On the benchmark $\mathbf{d}$ is the camera direction, so the row “GT direction” below is the published setting of [6, 5].

3. STEERABILITY: DEFINITION AND PROTOCOL

Rotational closure. First-order ambisonics are closed under rotation: for any $\mathbf{R} \in SO(3)$ the field seen in the rotated frame has $W' = W$ and directional channels $(X', Y', Z')^\top = \mathbf{R}(X, Y, Z)^\top$ (the sign convention of the Y channel is the corrected one of [6]), its energy map is $E'(\mathbf{u}) = E(\mathbf{R}^{-1}\mathbf{u})$, and a source at $\mathbf{d}$ moves to $\mathbf{R}\mathbf{d}$. The FoV video, being attached to the camera, is unchanged. A spatializer that uses $\mathbf{d}$ as a control should therefore be *equivariant*,

$$f(W, V, \mathbf{R}\mathbf{d}) \approx \mathbf{R} f(W, V, \mathbf{d}) \quad \forall \mathbf{R}, \quad (1)$$

which can be tested exactly, with the official metric, because both sides are computable from the data. Nothing in the training objective enforces (1): the data contain each clip in one frame only, and the camera directions cluster in front of and level with the camera (44% of test directions within 45° azimuth of the front, 90% within 30° elevation), so a model may satisfy the training loss by combining a weak reading of $\mathbf{d}$ with this prior. The equivariance test below uses rotations $\mathbf{R}_\theta$ about the vertical axis by θ , which preserve the elevation statistics of the data.

Protocol. All tests use the test split, the GT mono W , and per-clip random draws fixed by a seed. *(E) Equivariance:* draw $\theta \sim \mathcal{U}[0, 2\pi)$, feed $\mathbf{R}_\theta \mathbf{d}$, and score the output against $\mathbf{R}_\theta$ of the ground-truth FOA with CC/AUC. A perfectly steerable model scores as with the true direction; a model that ignores $\mathbf{d}$ scores near chance. *(F) Follow:* draw a uniform random unit target $\mathbf{t}$ (independent of the clip), feed it, and measure the angle between $\mathbf{t}$ and the peak of the predicted energy map on a 4° grid; we report the median and the fraction within 30° , overall and per octant of target azimuth and

band of elevation. The follow angle is also reported for every other condition, where it measures whether the output is where the *input* says. *(P) Precision:* rotate $\mathbf{d}$ by exactly $\delta \in \{10, \dots, 180\}^\circ$ about a random axis orthogonal to it and score against the true ground truth; the resulting curve $CC(\delta)$ tells how precise a user or a localizer must be, and where it crosses the score of a model trained without any direction. *Cheap controls:* $\mathbf{d}$ snapped to the nearest centre of a 7×14 equirectangular grid (cells of 25.7° , a click), a constant frontal direction, a random direction, the zero vector (no information), and the output of the automatic localizer of Sec. 4.

Rotation-consistency training. For every training clip we draw $\theta \sim \mathcal{U}[0, 2\pi)$ and replace the target by $\mathbf{R}_\theta$ of the ground-truth FOA and the input direction by $\mathbf{R}_\theta \mathbf{d}$, leaving W and V untouched. The augmented pair is exactly consistent, costs nothing, and removes the correlation between the direction and the prior, so the network can only satisfy the loss by reading $\mathbf{d}$. Variants: a uniform random $\mathbf{R} \in SO(3)$ instead of $\mathbf{R}_\theta$ (steerable in elevation too), rotating only half of the clips (keeping the original frame in the data), and additionally perturbing the input direction by a random angle $|\mathcal{N}(0, 15^\circ)|$ about a random orthogonal axis (*jitter*) to trade peak accuracy for tolerance. We verified the geometry numerically: after rotating clip and direction by any θ or $\mathbf{R}$, the direction falls within 30° of the peak of the rotated energy map for the same 82–85% of clips as without rotation (the remainder is the coarseness of the 7×14 check).

4. A LEAK-FREE PANORAMIC LOCALIZER AS AUTOMATIC CONTROL

To obtain a direction without the oracle we train a localizer on the full equirectangular frame. A frozen CLIP ViT-B/16 with positional embeddings interpolated to 14×28 patches encodes each 448×224 panorama at 2 fps; patch tokens pass through the final layer norm and projection (dense CLIP) and are pooled to a 7×14 grid of cells, each tagged with an MLP embedding of its unit direction (column-to-azimuth sign fixed once on training clips). A small convolutional encoder maps the log-magnitude STFT of W to one query per frame; multi-head attention from the query to the 98 cells yields an attention map $a_t \in \Delta^{98}$ whose expected direction $\hat{\mathbf{u}}_t = \sum_k a_{t,k} \mathbf{u}_k / \|\cdot\|$ is the localizer output. The attention is supervised by the ground-truth energy map decoded on the same 98 directions per 0.5 s window, $\mathcal{L}_{\text{loc}} = \frac{1}{T} \sum_t (1 - CC(a_t, g_t))$, a dense label that costs one 4×4 covariance per window; the attention context also conditions a spatializer end-to-end so that the localizer is trained jointly with the map loss, and the grid is rolled by k columns while the target field is rotated by $k \cdot 360^\circ / 14$ as augmentation. At test time the clip-mean of $\hat{\mathbf{u}}_t$ is fed to the steerable spatializer as $\mathbf{d}$.

5. EXPERIMENTS

Setup. Spatializers use AdamW (2×10^{-4} , weight decay 10^{-2}), batch 16, bf16, a budget of 7 h on one RTX 4090 (≈ 27 epochs), and the checkpoint with the best validation $CC_{\text{all}} + CC_{1\text{fps}}$ on a fixed 600-clip subset, all identical to [6]; the models differ only in the augmentation: none (the recipe of [6]), azimuth rotation consistency on every clip or on half of the clips, $SO(3)$ rotation consistency, and azimuth rotation with jitter. The localizer is trained for the same budget. All numbers are on the 6,553 test clips; 95% confidence intervals of differences are paired bootstraps over clips (2,000 resamples).

Table 1. Four spatializers driven by different direction inputs (6,553 test clips, ground-truth W , raw outputs). Follow: median angle ($^\circ$) between the peak of the predicted energy map and the *input* direction / fraction of clips within 30° . CC and AUC are full-clip scores against the ground truth, except row (E), which scores against the ground truth rotated by the same random θ as the input. For reference, the mean-map prior scores CC 0.374 and the FoV-only model of [6], trained without any direction, 0.500; a direction-trained model run with a zero vector scores 0.11–0.26.

Direction input	standard training [6]				+ azimuth rotation				+ $SO(3)$ rotation				+ azimuth rotation + jitter			
	CC	AUC	Fol.	< 30°	CC	AUC	Fol.	< 30°	CC	AUC	Fol.	< 30°	CC	AUC	Fol.	< 30°
GT direction (oracle)	0.783	0.914	9	0.94	0.797	0.918	8	0.97	0.805	0.930	3	1.00	0.786	0.914	9	0.96
snapped to 7×14 grid (a click)	0.766	0.907	8	0.94	0.780	0.910	8	0.97	0.791	0.923	3	1.00	0.779	0.911	10	0.96
panoramic localizer (Sec. 4)	0.326	0.667	11	0.96	0.316	0.654	13	0.96	0.343	0.686	3	1.00	0.390	0.690	19	0.75
constant frontal direction	0.332	0.666	6	0.99	0.334	0.660	7	0.96	0.336	0.678	4	1.00	0.392	0.690	13	0.86
random direction	0.064	0.535	13	0.91	0.017	0.506	10	0.98	0.049	0.530	3	1.00	0.155	0.572	26	0.56
(E) rotated input vs. rotated GT	0.742	0.897	12	0.87	0.801	0.921	7	0.97	0.805	0.930	3	1.00	0.785	0.914	9	0.96
(F) uniform random target	–	–	13	0.91	–	–	10	0.98	–	–	3	1.00	–	–	27	0.56

Table 2. Panoramic localizer variants (Sec. 4), trained end-to-end with a spatializer conditioned on the attention context (no oracle input). CC/AUC: full-clip scores of that end-to-end spatializer. Loc.: median angle ($^\circ$) between the predicted direction and the ground-truth peak per 0.5 s window / fraction within 30° .

Variant	CC	AUC	Loc. med.	< 30°
global CLIP token (no localizer)	0.466	0.747	–	–
dense tokens + audio-query attention	0.451	0.735	84.6	0.09
+ $\mathcal{L}_{loc}$	0.484	0.748	63.5	0.19
+ $\mathcal{L}_{loc}$ + grid-roll augmentation	0.409	0.715	60.7	0.22
visual-only query (no audio)	0.397	0.707	61.9	0.15
MSE only (no $\mathcal{L}_{map}$)	0.358	0.679	66.3	0.15
FoV CLIP, no direction [6]	0.500	0.752	–	–

5.1. Fidelity: does the model obey the direction?

Driven by the ground-truth direction, the model trained the standard way reaches CC 0.783 (Table 1), reproducing [6], and its energy peak lies within 30° of that direction in 94% of the clips. It is, however, only approximately steerable. When the direction is rotated and the output is scored against the rotated ground truth (row E), CC falls to 0.742 (-0.041 , 95% CI $[-0.046, -0.036]$) and the peak follows the input in 87% of the clips; a uniform random target (row F) is reached within 30° in 91% of the clips with a median error of 13° , with the misses concentrated at lateral targets (84–93% for $|\text{az}| \in [45^\circ, 135^\circ]$ vs. 97% in front; Fig. 2). Fig. 3 shows what this means: the peak moves with the input, but away from the frontal directions of the training data the field turns diffuse—right in position, wrong in shape. Rotation-consistency training removes this: row E equals the oracle row (0.801 vs. 0.797), random targets are followed in 98% of the clips (median 10°) uniformly over azimuth and elevation, and, as a faithful model must, a random direction yields chance-level CC (0.017) and a constant frontal direction 0.334, below the mean-map prior of 0.374 [6]. The $SO(3)$ variant goes further: every random target is reached within 30° with a median error of 3° , i.e. the control is exact at the resolution of the 4° peak grid, and row E equals the oracle row to three decimals. There is no price on the benchmark; the rotation-consistent models score *higher* with the true direction ($SO(3)$: $+0.022$ [0.018, 0.026], and $+0.030$ [0.026, 0.034] with a second training seed; azimuth: $+0.014$ and $+0.008$), which we attribute to the augmentation acting as a strong regularizer for an 8.4M-parameter network trained on 2,093 videos; rotating only half of the clips about the vertical axis gives 0.807 with

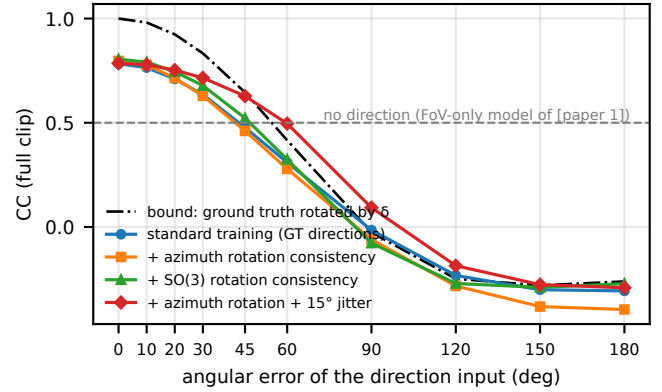


Fig. 1. Control-precision curve (P): full-clip CC when the direction input is the true direction rotated by exactly δ about a random orthogonal axis. Dash-dotted: the bound obtained by scoring the ground truth rotated by δ against itself; dashed: the FoV-only model of [6], i.e. no direction at all.

97% follow and an equivariance score of 0.796. Table 3 shows that every number above is reproduced by a second seed, that a trained standard model can be *retrofitted*: one hour of fine-tuning with $SO(3)$ consistency brings its follow rate to 100% (median 5°) and its equivariance score from 0.742 to 0.787 while adding $+0.010$ CC, and that a second family, a BiGRU mask estimator with the same conditioning, behaves the same way (follow $92\% \rightarrow 99\%$, equivariance $0.767 \rightarrow 0.809$, $+0.012$ CC [0.008, 0.017]). Jitter training is the instructive opposite: its random-target follow rate drops to 56% (79% for targets near the horizon, 28–38% above and below) and a random direction still scores 0.155, because a model trained on noisy directions learns to hedge between the input and the prior—it reintroduces exactly the behaviour that rotation consistency removes.

5.2. Tolerance: how precise must the direction be?

Fig. 1 traces $CC(\delta)$. The dash-dotted line is the geometric bound, the CC between the ground truth and itself rotated by δ (0.98 at 10° , 0.83 at 30° , 0.42 at 60° , chance at 90°): a perfectly faithful spatializer can do no better. The $SO(3)$ model follows this bound scaled by its own accuracy to within 0.01 at every δ ($0.805 \times 0.833 = 0.671$ predicted vs. 0.678 measured at 30° ; 0.335 vs. 0.324 at 60°), so for

Table 3. Robustness: two seeds, a one-hour $SO(3)$ retrofit of the trained standard model ($\text{lr } 5 \times 10^{-5}$), and a second family (BiGRU mask estimator, same conditioning, 17.4M parameters, 4h): CC with the true direction, under test (E), follow rate (F), and CC at $30^\circ / 60^\circ$ of direction error.

Model	seed	CC	CC (E)	(F) < 30°	$\delta=30^\circ$	$\delta=60^\circ$
standard	0	0.783	0.742	0.91	0.636	0.309
standard	1	0.784	0.740	0.88	0.637	0.320
+ azimuth rotation	0	0.797	0.801	0.98	0.629	0.279
+ azimuth rotation	1	0.791	0.787	0.96	0.631	0.300
+ $SO(3)$ rotation	0	0.805	0.805	1.00	0.678	0.324
+ $SO(3)$ rotation	1	0.813	0.808	1.00	0.682	0.326
standard $\rightarrow$ 1h $SO(3)$ retrofit	0	0.793	0.787	1.00	0.672	0.333
BiGRU, standard	0	0.799	0.767	0.92	0.650	0.342
BiGRU, $SO(3)$ rotation	0	0.811	0.809	0.99	0.682	0.323

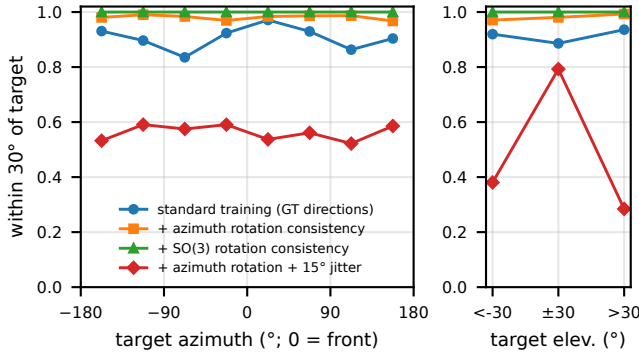


Fig. 2. Follow test (F): fraction of clips whose predicted peak lies within 30° of a uniform random target, by target azimuth (left; 0° is the front) and elevation (right).

a faithful spatializer the tolerance is a property of the metric and the data, not of the model: 10° of error costs 0.014 CC, 30° 0.127 and 60° 0.48, and at $\delta^* \approx 47^\circ$ the curve crosses the FoV-only model of [6] (0.500, dashed), beyond which a direction is worse than none. The standard and the azimuth-only model fall 0.02–0.05 short of their own products at 30 – 45° , i.e. the law holds to the degree that the model is faithful, and cross at $\delta^* \approx 42^\circ$; the jitter model degrades more gracefully ($\delta^* \approx 60^\circ$; +0.19 CC over the standard model at 60°), but only by hedging, which is the loss of fidelity measured above: tolerance and fidelity are traded against each other by the noise seen in training. Snapping the direction to the 7×14 grid costs 0.015–0.017 CC for the faithful models ($SO(3)$: [0.014, 0.016]), so a click on the panorama retains 95% of the oracle gain over no direction.

5.3. Automatic control from the panorama

Table 2 ablates the localizer. Without $\mathcal{L}_{\text{loc}}$ the attention is not a localizer (median error 84.6° , chance is 90°); the free dense label brings it to 63.5° and the grid-roll augmentation to 60.7° with 22% of windows within 30° (25% for the clip-mean direction). The audio query matters little for localization (61.9° with a visual-only query) but does for the spatializer, and $\mathcal{L}_{\text{map}}$ remains essential. The end-to-end CC of every panoramic model stays at or below the FoV-only model (0.500), and the augmentation that improves localization lowers CC (0.484 to 0.409): the metric rewards the frontal prior that the augmentation removes, and the shrinkage factor of [6] drops from 0.33 to 0.10, i.e. the model, unsure where the source is, spreads its en-

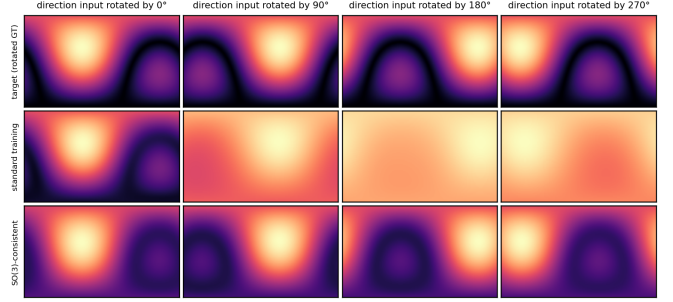


Fig. 3. Steering one test clip: whole-clip energy maps (equirectangular, azimuth $-180 \dots 180^\circ$, each normalized to its maximum) of the ground truth rotated by $0/90/180/270^\circ$ (top) and of the standard (middle) and $SO(3)$ -consistent (bottom) spatializers driven by the correspondingly rotated direction. The standard model moves its peak but loses the contrast of the field away from the front.

ergy. Feeding the localizer’s direction to the spatializers (“localize, then steer”, Table 1) gives CC 0.32–0.34 for the faithful models and 0.39 for the jitter model: below the end-to-end models and far below the FoV-only baseline (0.500), and slightly above what the precision curve predicts when averaged over the localizer’s per-clip error distribution (0.24–0.27 and 0.36; the curve perturbs directions isotropically, whereas the localizer errs mostly toward the front, where the prior lies). A faithful spatializer puts the sound exactly where a localizer with 58° median clip error says, and Fig. 1 says that this is worse than saying nothing; automatic control needs a localizer whose errors stay below δ^* , i.e. roughly 40 – 45° . The localizer is the bottleneck, and its errors are not uniform: for the 49% of test clips whose true peak lies within 45° azimuth of the front, the median error is 30.7° (49% within 30°), whereas sources to the side or behind are essentially never found (medians 80.9° and 140.8° , 2% and 0% within 30°). The grid-roll augmentation did not remove this prior but taught the model where the front is (the operator and the stitching seam are visible, and the dominant sound is very often the person filming). Peak ground-truth maps are easier (median 48° vs. 62° for the most diffuse third), pointing at a second cause: ambient fields whose arg-max is not a visible object.

6. CONCLUSION

The direction input of a video-to-ambisonics spatializer is its interface to users, trackers and localizers, and the exact rotational closure of first-order ambisonics makes its behaviour testable without any annotation; we recommend reporting the equivariance and follow tests next to any benchmark score, since the oracle setting cannot tell a controllable model from one that memorised the prior. Tested this way, the standard recipe of training on ground-truth directions is only approximately steerable—right in position, wrong in shape away from the frontal directions of the data—whereas $SO(3)$ rotation-consistency training, at zero cost and also as a one-hour retrofit, makes the control exact and raises the benchmark score, across seeds and architectures. The precision curve of a faithful model is a data-only bound scaled by its accuracy: 30° of error costs 0.13 CC, a click on a coarse grid costs 0.015, and beyond $\approx 47^\circ$ a direction is worse than none, a threshold our leak-free panoramic localizer (58° median error) does not meet; noisy-direction training raises it only by hedging. Limitations: one benchmark, energy-map metrics only, single-direction tests.

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